

- 1 Fig. 1.1 is a diagram showing the structure of a section of a DNA molecule.

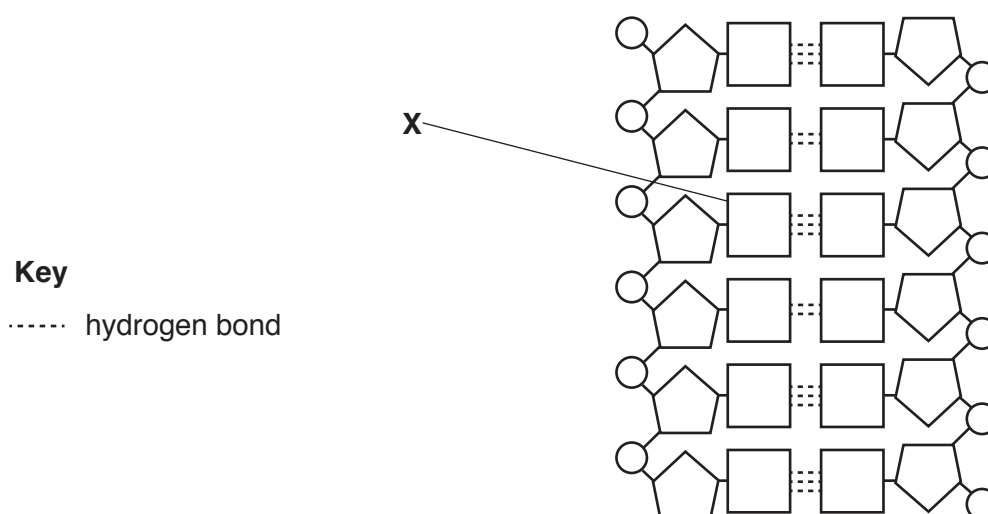


Fig. 1.1

- (a) Draw a circle around **one** monomer of DNA in Fig. 1.1. [1]

- (b) Name the two bases forming the base pair at **X** in Fig. 1.1 **and** give a reason for your answer.

bases

reason

.....

.....

..... [2]

- (c) The statements 1–5 describe events that occur during DNA replication.

- 1 DNA polymerase forms a phosphodiester bond
- 2 DNA double helix forms
- 3 hydrogen bonds break
- 4 hydrogen bonds form
- 5 two strands of the double helix separate

Write the numbers 1 to 5 in the spaces below to show the order in which these events occur.
 The first one has been done for you.

3 [1]

(d) The telomere is a region found at the end of a chromosome.

Outline the function of telomeres.

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..... [2]

[Total: 6]